
Inside Model Analysis

How the parameters are computed, and how the engine is verified

A methodology companion to the NEXUS-1 console

What this is	A precise account of what the <i>Model Analysis</i> screen does: the equations the engine solves, the numerical scheme it uses, and — the heart of the page — how each figure is checked against an <i>independent</i> reference. Written for an engineer who wants to explain the page, not just show it.
The one word that matters	This page performs <i>verification</i> (is the solver solving the equations correctly?), not <i>validation</i> (does the model match a real reactor?). The distinction is made explicit in §1 and carried throughout.
Ground truth	Every method described here is read from the running solver. Formulas in this document are the formulas in the code; nothing is illustrative unless labelled so.
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The claim this page makes, stated plainly. “The solver reproduces the closed-form analytical solutions of the point-kinetics equations to within a stated numerical tolerance, and every constant shown is the constant the simulation actually runs.” **The claim it does not make:** that the numbers match any specific real reactor. The constants are representative; agreement with plant data is not asserted.

1 Verification vs. validation — what the page actually claims

In simulation engineering these two words are not synonyms, and keeping them apart is what makes a model defensible.

- **Verification** asks “are we solving the equations right?” — is the numerical scheme faithful to the mathematics it claims to integrate? You answer it by comparing the solver’s output to an *independent* solution of the same equations: a closed-form analytical result, a conservation law, or a known limiting case.
- **Validation** asks “are we solving the right equations?” — does the model, however well solved, actually describe the physical reactor? You answer it only with measured data from the real system.

The Model Analysis screen does **verification**. For a step reactivity insertion it computes the reactor period two completely independent ways — once from the analytical *inhour equation* and once by running the solver — and reports their disagreement. If the solver is integrating correctly, the two must coincide. That is a statement about *numerical correctness*, and it is true regardless of which reactor (if any) the constants describe.

It deliberately does **not** perform validation. The constants are representative textbook-order values, chosen per unit class; the page never claims they reproduce a measured plant. This is stated on the screen itself and is the honest position: a portfolio demonstrator has no licensed plant data to validate against, so it verifies what it can prove and is explicit about the rest.

Why this matters for a demo. A reviewer who knows reactors will ask “how do you know this is right?” The correct answer is not “it matches a real PWR” (it doesn’t claim to) but “the solver reproduces the analytical solutions of the equations to a stated tolerance, and here are the checks.” That answer is both true and the stronger one.

2 The equations being solved

The engine integrates the standard **point-kinetics equations** with six delayed-neutron precursor groups, coupled to a lumped thermal model and an iodine/xenon poison model. With n the neutron population (relative to rated), C_i the precursor concentrations, Λ the prompt generation time, β_i the delayed fractions ($\beta = \sum_i \beta_i$) and λ_i their decay constants:

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^6 \lambda_i C_i, \quad \frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i. \quad (1)$$

Reactivity ρ (in pcm; 1 pcm = 10^{-5}) is the sum of control-rod worth and three feedbacks:

$$\rho = \rho_{\text{rod}} + \underbrace{\alpha_D (T_F - T_{F0})}_{\text{Doppler, fuel}} + \underbrace{\alpha_M (T_C - T_{C0})}_{\text{moderator, coolant}} - \underbrace{K_{Xe} (Xe - Xe_{eq})}_{\text{xenon-135}}. \quad (2)$$

Temperatures follow a two-node lumped model (fuel node T_F , coolant node T_C , inlet T_{in} , power $P \equiv n$):

$$C_F \frac{dT_F}{dt} = P K_P - K_{FC} (T_F - T_C), \quad C_C \frac{dT_C}{dt} = K_{FC} (T_F - T_C) - K_{CO} (T_C - T_{in}). \quad (3)$$

Iodine-135 and xenon-135 obey the usual coupled balance, with γ the fission yields and $\sigma\varphi$ a lumped burnout term:

$$\frac{dI}{dt} = \gamma_I \text{SF} P - \lambda_I I, \quad \frac{dXe}{dt} = \gamma_{Xe} \text{SF} P + \lambda_I I - \lambda_{Xe} Xe - \sigma\varphi P Xe. \quad (4)$$

These are not simplified stand-ins — they are the equations whose textbook solutions the verification step checks against.

3 How the solver integrates them

Point kinetics is *stiff*: the prompt timescale ($\Lambda \sim 10^{-5}$ s) and the slowest precursor ($\lambda_1 \sim 0.012 \text{ s}^{-1}$, i.e. ~ 80 s) differ by seven orders of magnitude. A naive explicit step would explode unless Δt were impractically tiny. The engine uses **operator splitting** with a stability-preserving scheme for each part.

3.1 Precursors — exact analytic update

Over a step Δt , holding n constant, each precursor equation is linear and integrates in closed form:

$$C_i^{k+1} = C_i^k e^{-\lambda_i \Delta t} + \frac{\beta_i n}{\Lambda \lambda_i} (1 - e^{-\lambda_i \Delta t}). \quad (5)$$

This is unconditionally stable and exact for constant n — there is no truncation error in the precursor decay.

3.2 Neutrons — implicit (backward) Euler

With the delayed source $S = \sum_i \lambda_i C_i^{k+1}$, the neutron population uses a backward-Euler step, which stays stable through the stiff prompt term:

$$n^{k+1} = \frac{n^k + \Delta t S}{1 - \Delta t (\rho - \beta)/\Lambda}. \quad (6)$$

For update stability near prompt criticality the reactivity used in this term is clamped at $\rho \leq 0.95 \beta$ (the prompt-critical pole is approached but not crossed numerically). Fuel/coolant temperatures and the iodine/xenon balance, being slow, advance with explicit Euler. Each visible frame is divided into sub-steps of ≤ 0.02 s.

Single source of truth. The verification checks call the *same* step function the live simulation uses — not a re-implementation. So a check cannot “pass” against code that differs from what the console runs. The parameters panel, the JSON export and the running model are provably the same numbers.

4 The verification checks — principle and method

The principle is identical for every row: *compute a quantity two independent ways — once analytically, once by running the solver — and report the discrepancy.* A small discrepancy is evidence the integration is faithful; a large one would expose a bug. The table below is what the *Solver Validation* panel computes live.

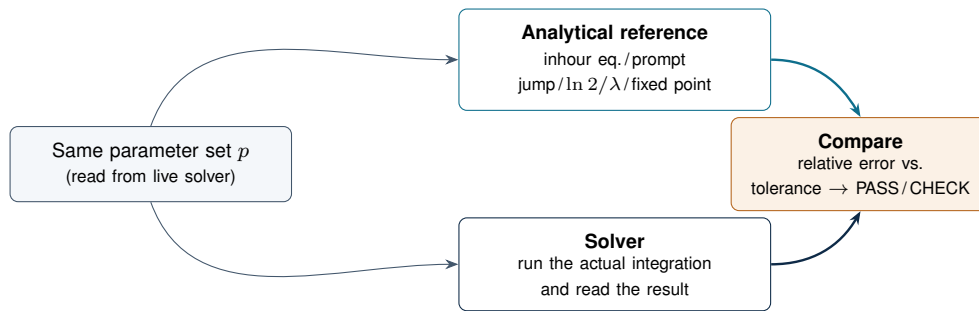


Figure 1: The verification pattern, applied to every check on the page. The two paths share one parameter set but are otherwise independent: one is a hand-derived closed form, the other is the running solver. Their agreement is the evidence.

4.1 Asymptotic reactor period — the central check

For a step insertion ρ , the stable period $T = 1/\omega$ where ω is the dominant root of the **inhour equation**:

$$\rho = \omega \Lambda + \sum_{i=1}^6 \frac{\beta_i \omega}{\omega + \lambda_i}. \quad (7)$$

The reference solves this for ω by bisection. The “measured” value comes from the *solver*: it runs the bare kinetics (feedback off) forward, then fits a straight line to $\ln n(t)$ over the asymptotic window (300–400 s) whose slope is $1/T$. The page reports $|T_{\text{solver}} - T_{\text{inhour}}|/T_{\text{inhour}}$ for +10 and +100 pcm; agreement to $< 0.5\%$ passes. This is the strongest statement on the page — two independent routes to the same dynamic quantity.

4.2 Prompt jump

Immediately after a sub-prompt insertion, the neutron level jumps by the analytical factor

$$\frac{n_+}{n_0} \approx \frac{\beta}{\beta - \rho}. \quad (8)$$

The solver is stepped 0.05 s from equilibrium at +200 pcm and compared to this. It checks the fast (prompt) response specifically.

4.3 Delayed fraction, decay half-lives, xenon-pit timing

- **β (six-group sum)** defines the dollar: $\rho_{\$} = \rho/\beta$, so $\beta \equiv \$1.00$. Reported as a data check.
- **Half-lives** $t_{1/2} = \ln 2/\lambda$ for I-135 and Xe-135 are compared to textbook ≈ 6.57 h and ≈ 9.14 h — confirming the decay constants are physically correct.
- **Post-shutdown xenon peak**: from 100 % equilibrium the source is removed and I/Xe integrated forward; the peak should fall in the classic ≈ 7 –11 h window. A qualitative range check that the famous “xenon pit” timing emerges.

4.4 Steady-state null test and thermal self-consistency

Two internal-consistency checks confirm the operating point is a genuine fixed point:

- **Steady state** ($\rho = 0$): the full model (feedback on) is run 60 s from the 100 % equilibrium with rods at zero; the maximum $|n - 1|$ must stay below 0.05%. Any spurious drift would reveal an integration fault.
- **Thermal equilibrium**: the temperatures the thermal model implies at full power,

$$T_C^{eq} = T_{in} + \frac{K_P}{K_{CO}}, \quad T_F^{eq} = T_C^{eq} + \frac{K_P}{K_{FC}}, \quad (9)$$

are compared to the model's own reference temperatures T_{C0}, T_{F0} . They must match — which is exactly why the reference temperatures are *derived* from the thermal gains per unit, not set independently.

5 Where the parameters come from

The *Parameters* panel reads the live solver and separates two honestly different categories:

- **Nuclear data (physical constants)** — the six-group β_i, λ_i (U-235 thermal-fission evaluated data), the I-135/Xe-135 decay constants and fission yields. These are physical and are not “chosen.”
- **Model parameters (representative choices)** — the generation time Λ , feedback coefficients α_D, α_M , lumped xenon worth, the thermal gains K_P, K_{FC}, K_{CO} , node heat capacities, and the reference operating temperatures. These are representative textbook-order values for the unit class, not measured.

Per unit class the representative set changes (e.g. a boron-free integral SMR is given a stronger moderator coefficient and cooler operating temperatures), which is why the period, equilibrium temperatures and xenon behaviour differ when you switch units. The labelling on-screen always states “representative constants for this unit class.”

6 Sensitivity and uncertainty

6.1 Normalized sensitivity (central difference)

For each output metric Y and parameter p , the page reports the dimensionless logarithmic sensitivity, estimated by a $\pm 1\%$ central difference:

$$S_p = \frac{p}{Y} \frac{\partial Y}{\partial p} \approx \frac{Y(p(1+h)) - Y(p(1-h))}{2h Y_0}, \quad h = 0.01. \quad (10)$$

Because S_p is dimensionless it is legitimately comparable across parameters. The metrics are the asymptotic period, the equilibrium fuel temperature, and the post-shutdown xenon-pit depth. A deliberate teaching point: the *dominant* parameter differs by output — Λ and β drive the period, the thermal gains drive temperature, the xenon constants drive the pit — so there is no single “most influential parameter.”

6.2 Propagated (model-form) uncertainty

First-order propagation gives the output band from assumed input spreads:

$$\sigma_Y^2 \approx \sum_i \left(\frac{\partial Y}{\partial p_i} \right)^2 \sigma_{p_i}^2, \quad \text{band} = 2\sigma_Y. \quad (11)$$

The input spreads are *illustrative* (nuclear data $\pm 2\%$, modelling parameters $\pm 10\text{--}15\%$). This captures *model-form* uncertainty only — it is explicitly distinct from sensor or measurement noise, which a point-kinetics model does not represent.

7 Live margins and the xenon forecast

Two panels read the *current* solver state rather than a closed form:

- **Operating margins** compare live power, period, fuel temperature and reactivity against demonstrator setpoints (118 % overpower, 10 s short-period, a representative fuel-temperature band, and \$1.00 prompt critical). These are illustrative setpoints, *not* real Technical Specification limits.
- **Xenon forecast** integrates the I/Xe equations forward from the current state with power held constant, reporting the projected peak and its timing — a genuine forward integration, not a linear extrapolation, so it captures the rise-turnover-decay shape.

What the page refuses to show, and why. DNBR and linear heat rate are intentionally absent. They are thermal-hydraulic outputs requiring a subchannel-flow model, a critical-heat-flux correlation and an axial power distribution. A point-kinetics model with a single lumped thermal node cannot compute them, so displaying a number would be fabrication. Declining to show them is itself a correctness statement.

8 Honest scope — what this verifies and what it does not

The page does establish	The page does not establish
The solver reproduces the analytical solutions of the point-kinetics equations to a stated tolerance.	That the constants match any specific real reactor (no plant data is used).
The displayed parameters are exactly those the simulation runs.	That the lumped thermal/xenon model captures spatial or thermal-hydraulic detail.
The qualitative phenomena (prompt jump, period, xenon pit, self-regulation) emerge correctly.	DNBR, linear heat rate, or any subchannel quantity.
The model-form sensitivity and uncertainty structure.	Measurement / sensor uncertainty.

In one line: *this is verified, representative physics — the equations are solved correctly and transparently, and the document is careful to claim nothing more.*

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